

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Code of Conduct:

I B. Chaitanya Prakash Reddy (192425255)
(Name / Reg No) certify that this submission is my original work and
that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Chaitanya
Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%				
Debugging	25%				
Optimization	20%				
Analysis	20%				
Code Quality	15%				

Chaitanya
Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

C02 - AT3

Q1.

The workstation is configured with IP Address 192.168.10.65, subnet mask 255.255.255.192 (/26) and default gateway 192.168.10.1. The system cannot communicate because the IP address and the default gateway belong to different subnets.

- Subnet: 192.168.10.64/26.
- Network Address: 192.168.10.64.
- Broadcast Address: 192.168.10.127
- Valid Host Range: 192.168.10.65 - 192.168.10.126
- Error: The default gateway (192.168.10.1) belongs to the 192.168.10.0/26 subnet, so it is not valid for the configured host.
- Optimised configuration: Change the IP address to 192.168.10.10 and use 192.168.10.1 as the default gateway.
- Usable Host Addresses: 62.

Q2.

The two devices are configured with different IPv6 addresses. Communication fails because they do not belong to the same subnet. The IPv6 configuration must be corrected so that both devices use the same /64 network prefix.

- Device A : $2001:db8:0000:0000:0000:ff00:0002:8329$
- Device B : $2001:db8:0000:0000:0001:0000:0000:0001$
- Network prefix of A : $2001:db8:0:0::/64$
- Network prefix of B : $2001:db8:0:0::1::/64$
- Error : The devices are in different $/64$ subnets, so communication is not possible.
- Optimized configuration : Assign Device B an address such as $2001:db8:0:0::2/64$, so both devices are in the same subnet.
- Available Host Addresses : 2^{64} (approximately 18.4 quintillion addresses).

Q3.

The network $192.168.1.0/24$ is divided using $/26$, $/27$, and $/28$ subnets. Some departments have unused IP addresses; while others do not have enough. The subnet allocation should be analyzed and optimized using VLSM for better utilization.

- $/26$: 62 usable host addresses
- $/27$: 30 usable host addresses
- $/28$: 14 usable host addresses

- Issue: some departments are assigned larger subnets than required, and leading to wasted IP addresses, while others have insufficient addresses.
- Debugging: Check the network address, broadcast address, and valid host range for each subnet to identify allocation mistakes.
- optimized solution: Use VLSM to assign to subnet sizes based on the actual number of hosts required.
- Result: VLSM reduces IP address wastage and improves overall network efficiency.

Q4.

The two devices have different IPv6 network prefix, so they cannot communicate even though they are connected to the same network. Both devices should be configured with the same /64 prefix for successful communication.

- Device A: 2001:0db8:0000:0000:0000:ff00:0042:8329.
- Device B: 2001:0db8:0000:0000:0001:0000:0000:0001
- Issue: The device belong to different /64 subnets.
- optimized configuration: Assign Device B an address such as 2001:db8:0:0::2::1/64 so that both devices are in the same subnet.

- Available Host Addresses: 2^{64} (approximately 18.4 quintillion usable addresses).

Q5.

The router contains routes for $192.168.1.0/24$, $192.168.2.0/24$ and a default route. Packets for $192.168.2.100$ should match the correct Routing entry using the longest-prefix-rule. The routing table should be checked for errors and optimized if required.

- Matching Route: $192.168.2.0/24$
- Next-Hop Router: $10.0.0.3$
- Verification: The destination $192.168.2.100$ belongs to the $192.168.2.0/24$ network.
- Routing error: verify that the next-hop router and routing table entries are correctly configured.
- Optimized solution: keep accurate routing entries and remove any incorrect or duplicate routes to improve forwarding efficiency.
- If no matching route exists: The router uses the Default Route ($10.0.0.1$).